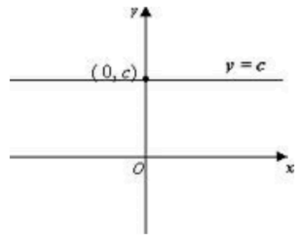
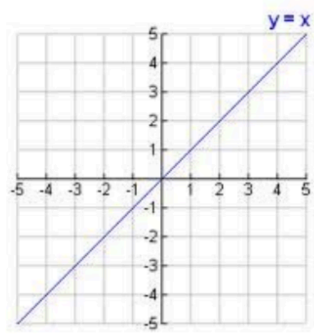
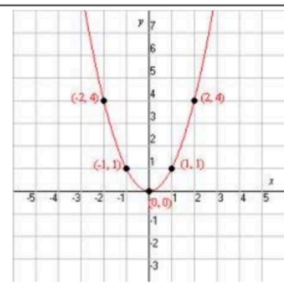
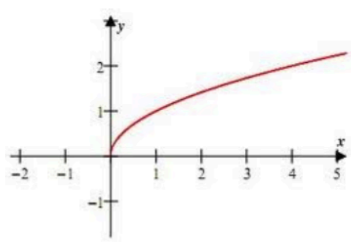
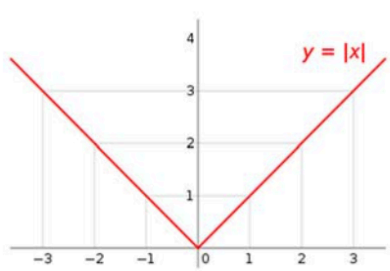
Parent Functions Cheat Sheet

Function Name	Parent Function	Graph	Characteristics
Algebra			
Constant	$f(x) = c$		Domain: $(-\infty, \infty)$ Range: [c, c] Inverse Function: Undefined (asymptote) Restrictions: c is a real number Odd/Even: Even General Form: $Ay + B = 0$
Linear or Identity	$f(x) = x$		Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$ Inverse Function: $g(x) = x$ Restrictions: m ≠ 0 Odd/Even: Odd General Forms: $Ax + By + C = 0$ $y = mx + b$ $y - y_0 = m(x - x_0)$
Quadratic or Square	$f(x) = x^2$		Domain: $(-\infty, \infty)$ Range: $[0, \infty)$ Inverse Function: $g(x) = \sqrt{x}$ Restrictions: None Odd/Even: Even General Form: $Ax^2 + By + Cx + D = 0$
Square Root	$f(x) = \sqrt{x}$		Domain: $[0, \infty)$ Range: $[0, \infty)$ Inverse Function: $g(x) = x^2$ Restrictions: $x \geq 0$ Odd/Even: Neither General Form: $f(x) = a\sqrt{b(x-h)} + k$
Absolute Value	$f(x) = \|x\|$		Domain: $(-\infty, \infty)$ Range: $[0, \infty)$ Inverse Function: $f(x) = x \; for \; x \geq 0$ Restrictions: $f(x) = \begin{cases} x, & if \; x \geq 0 \\ -x, & if \; x < 0 \end{cases}$ Odd/Even: Even General Form: $f(x) = a\|b(x-h)\| + k$

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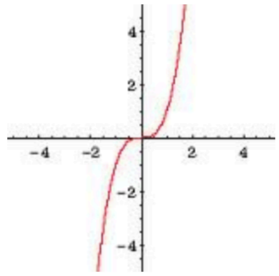
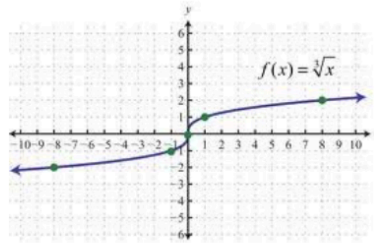
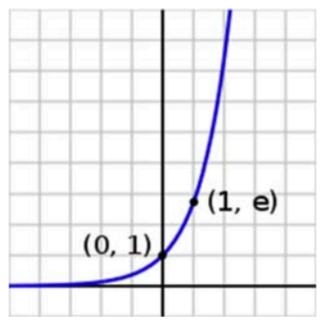
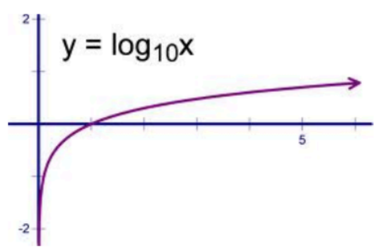
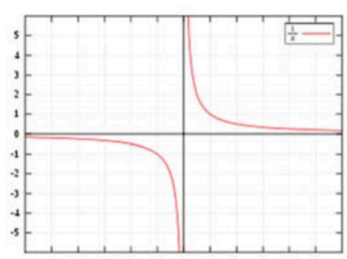
Cubic	$f(x) = x^3$		Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$ Inverse Function: $g(x) = \sqrt[3]{x}$ Restrictions: None Odd/Even: Odd General Form: $f(x) = a(b(x-h))^3 + k$
Cube Root	$f(x) = \sqrt[3]{x}$		Domain: $(-\infty, \infty)$ Range: $(-\infty, \infty)$ Inverse Function: $g(x) = x^3$ Restrictions: None Odd/Even: Odd General Form: $f(x) = a\sqrt[3]{b(x-h)} + k$
Exponential	$f(x) = 10^x$ *or* $f(x) = e^x$		Domain: $(-\infty, \infty)$ Range: $(0, \infty)$ Inverse Function: $g(x) = \log x$ *or* $g(x) = \ln x$ Restrictions: None, x can be imaginary Odd/Even: Neither General Form: $f(x) = a\,10^{(b(x-h))} + k$
Logarithmic	$f(x) = \log x$ *or* $f(x) = \ln x$		Domain: $(0, \infty)$ Range: $(-\infty, \infty)$ Inverse Function: $g(x) = 10^x$ *or* $g(x) = e^x$ Restrictions: $x > 0$ Odd/Even: Neither General Form: $f(x) = a\log(b(x-h)) + k$
Reciprocal **or** **Rational**	$f(x) = \frac{1}{x}$		Domain: $(-\infty, 0) \cup (0, \infty)$ Range: $(-\infty, 0) \cup (0, \infty)$ Inverse Function: $g(x) = \frac{1}{x}$ Restrictions: $x \neq 0$ Odd/Even: Odd General Form: $f(x) = a\left[\frac{b}{(x-h)}\right] + k$